

SAD

Information Gathering and Interactive Method

Interview: directed convo with a specific purpose, in a QnA format

Goal: understand perspectives. → current state

→ org. & personal goals.
→ HCT.

Focus on opinions. Uncover goals

Step zero: Interview yourself and ~~unders~~ visualize the interview
Recognize biases.

Steps of Interview Preparation:

- 1) Read bg → understand the org.
→ maximise interview time (not asking available ques)
- 2) ~~Step~~ Establish interviewing objectives → Identify key areas
→ HCT
→ DBMS.
- 3) Decide the interviewee → include people at all levels
→ maximise diversity
→ consult your primary contact.

- 4) Prepare the Interviewee
- Inform them
 - format / location
 - time limit: (45-60 mins)

- 5) Planning Questions
- Open Ended
 - Closed
 - Probes

Open Ended Questions

Pros

- a) Put interviewee at ease
- b) Reveals attitude
- c) Interesting
- d) Spontaneous

Cons

- a) irrelevance risk
- b) time consuming
- c) can seem unprepared.
- d) interviewer may lose control

Closed Questions

Pros

- a) saves time
- b) yields data
- c) easy comparison
- d) Interviewer keeps control

Cons

- a) boring
- b) lacks detail
- c) might miss real issue
- d) less rapport building.

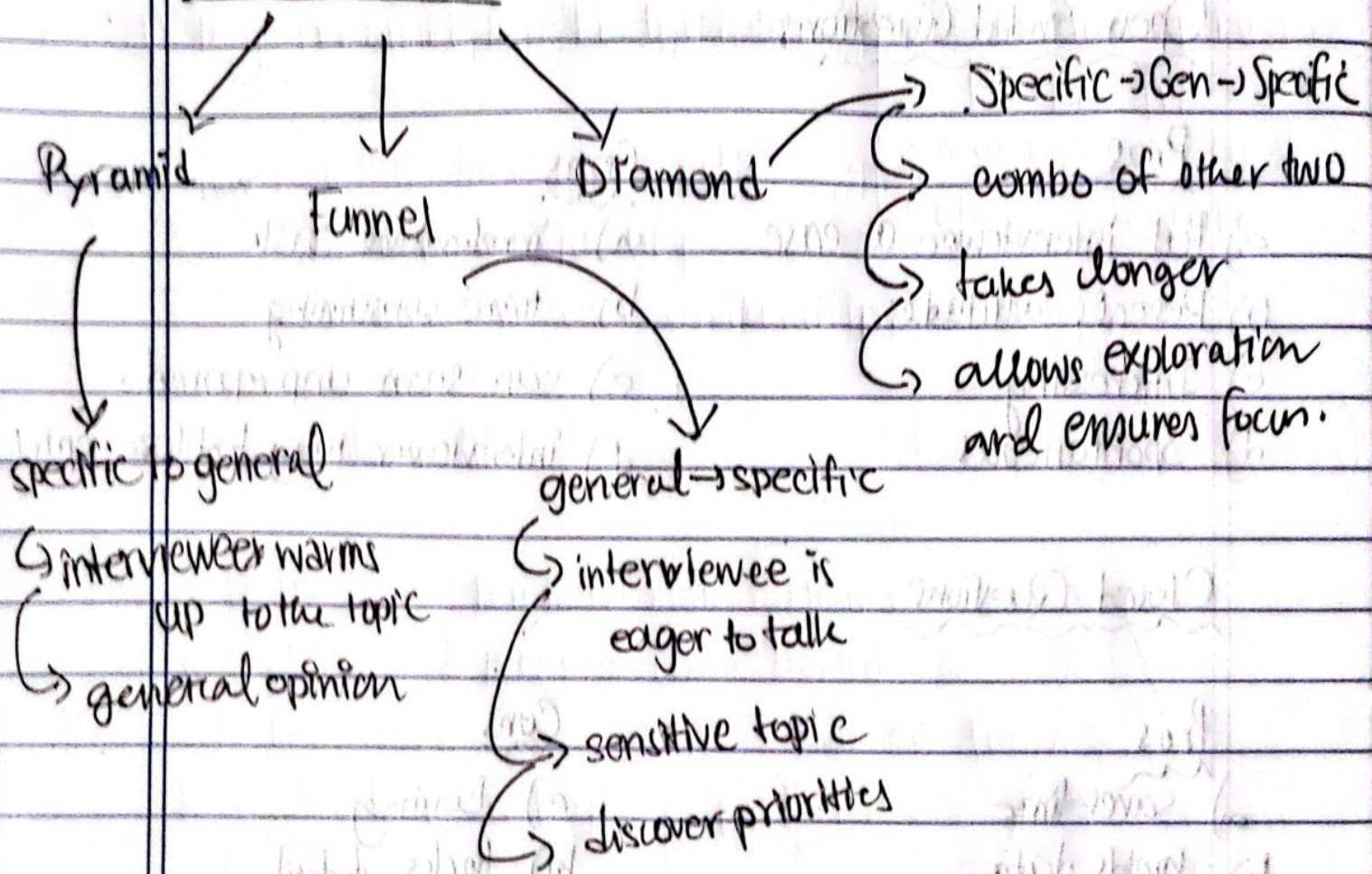
doesn't help build conf.
trust, connection etc.

* Open ended questions give depth but are harder to analyze
Closed questions give precise, analyzable data but lacks richness

Probe Questions

Follow-up questions to get more clarity, meaning and details.

Interview Flow



Ending: "any else important?"

"do else do you rec to be interviewed?"

"thanks"

After Interview: 1) ~~R~~ Write report
2) follow-up

Stories:

- Rich details
- Organizational history, shared info & values
- enduring stories

Narrative:

- understand past experiences
- less time efficient, requires patience
- more memorable, insightful, reveal behaviors.

→ Trigger

→ Goal

→ Problems (Obstacles)

→ Transformation

→ Result

→ Learned lessons.

→ aftermaths.

JAD: Joint Application Design

Structured group ~~work~~ workshop bringing analysts, users, and executives together to define requirements and design interfaces jointly. ~~Use~~ Can be used when:

a) new, innovative solutions are needed

b) eager, ~~is~~ supports collab.

c) needs diverse ideas

d) key personnel can be freed up

→ users

→ executive sponsor

→ IS Analyst

→ session leader (neutral)

→ Observers & Scribe

Pros: a) cut time and cost

b) improve quality

c) increase user ownership.

Settings for JAD

- Offsite → minimizes distractions
- projectors + laptops + white board
- schedule when everyone can commit 100%.
- face-to-face is highly preferred.

Cons: a) Hard to schedule 2-4 full uninterrupted days

b) inadequate planning wastes time

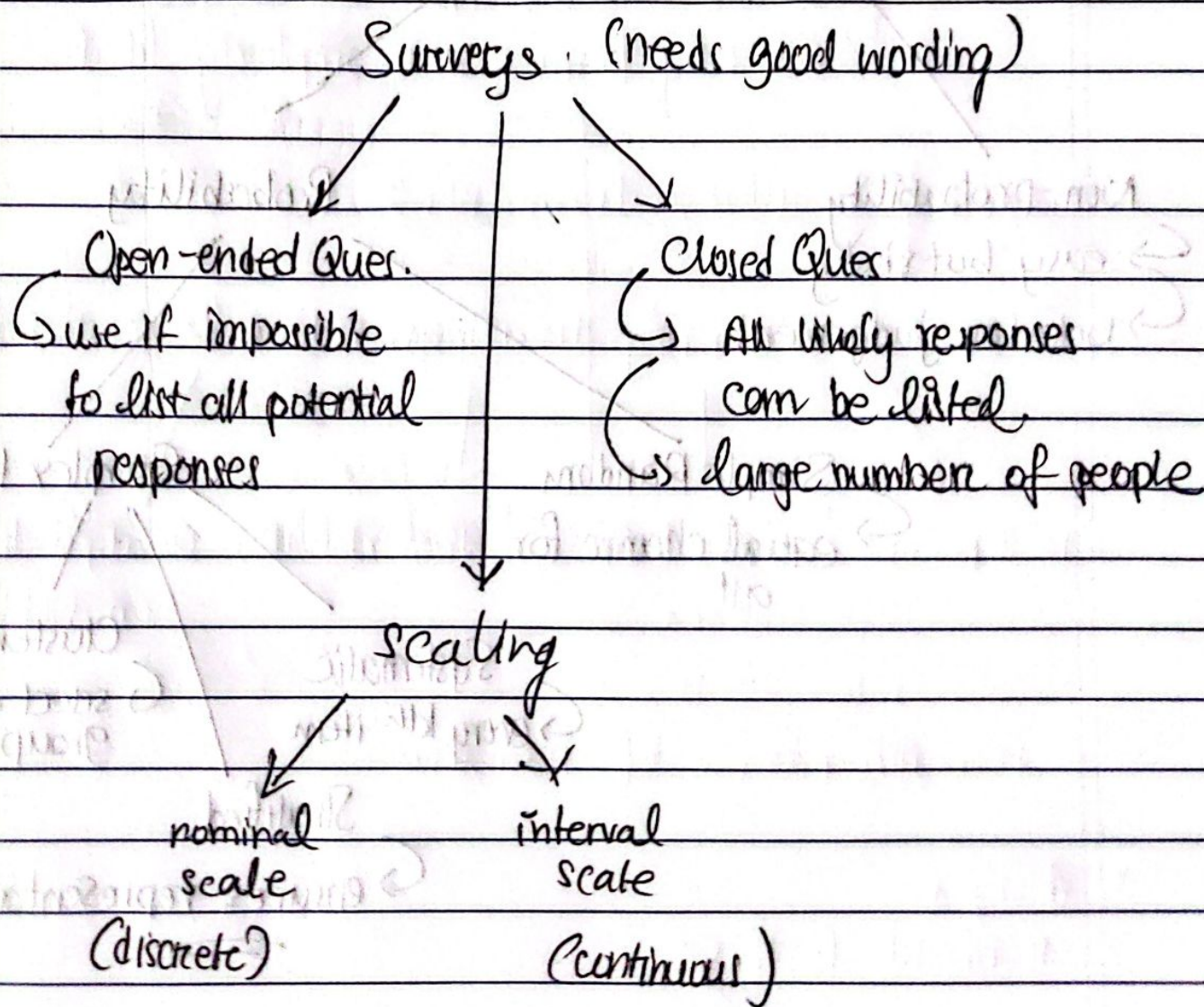
c) poor ~~data~~ documentation = lost value

d) less predictable than interviews.

e) requires intense collaboration.

Questionnaires:

- Useful for quantifying responses using closed questions.
- Can complement interviews:
 - a) clarify questionnaire questions with interviews.
- Useful if:
 - a) people are widely dispersed.
 - b) Need input from many.
 - c) doing exploratory study to gauge overall opinion early on.
 - d) want to identify common problems.



Information Gathering : Unobtrusive Methods

Sampling : systematically select representative elements of a population. Why? → examining everything is expensive

- speed ↑
- deeper analysis on small set
- avoid skewed perspectives

Types of Samples

Non-probability
→ easy but risky
→ based on judgement

Probability

Simple Random
→ equal chance for all

Systematic
→ every kth item

Stratified
→ ensures representation

Complex Random

Cluster
→ select a whole group.

Sample Size Formula

To find out how many samples we should take, we have to first set some parameters.

- param 1: ^{guess proportion} est. error rate (p) $\rightarrow$ 0.5 if unsure
- param 2: acceptable error (i)
- param 3: confidence level $\rightarrow$ lookup corr. z value

Standard error, $\sigma_p = \sqrt{p(1-p)}$

Sample size, $n = \frac{p(1-p)}{\sigma_p^2} + 1$

BUT, a well designed sample can reduce required sample size but still give represent the whole set. Quality > Quantity.

Analyzing Quantitative Does

- ① Compare Actual vs Intended → find trends and find the reason
- ② Check records ~~and try~~ → try to find errors
 - bad design?
 - automation possible?
- ③ Find patterns → Good or Bad
- ④ Forms →
 - collect all forms → official & bootleg
 - understand the input and output
 - ask why forms are incompletely filled

Decoding Qualitative Documents :

Focus on uncovering meanings, values & relationships

↓ ↓ ↓ ↓

metaphors "us vs them" "good vs evil" analyze themes on websites

→ Recognize presence of humour.

- design
- technical factors

Analyzing Unstructured Data

Use Text Analytics

→ using softwares to analyze large ~~data~~ amounts of text and find patterns.

general process

- 1) Input documents
- 2) Clean text (ex: remove stop words)
- 3) Analysis → the software counts keywords and ^{words} ~~work~~ that appear together often. (login, password...)
(login prob.)
- 4) Output
 - ├ Summaries
 - ├ ranked words
 - ├ themes
 - └ visual diagrams

Interpret the Leximancer results to get insights and develop recommendations and improvements.

Decision Makers

Observe decision makers and their ~~interest~~ interactions with environment and computers, their relationships with others. Managerial work is often fragmented - difficult to capture via interviews.

Playscript

Decision Maker	Actions (verbs)

- helps determine info requirement for decisions
- shows dependencies
- shows info sources

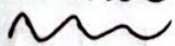
STROBE: Structured Observation of the Environment

Structured research method used to record specific, predetermined behaviours or physical elements within a setting.

7 elements:

- 1) Office location : corner offices, accessibility
- 2) Desk placement : encourage / create a barrier
- 3) Stationary Equipment : large filing cabinet - personal storage
- 4) Props : PC, smartphones
- 5) Office Lighting
- 6) Clothing
- 7) External Info Sources : journals, newspapers, books

Method : Anecdotal Checklist



- identify themes from 7 elements
- compare observation to the narratives
- use shorthand symbols on a checklist.
 - ✓ : confirms the narrative
 - ✗ : negates " "
 - : investigate further
 - : modifies the narrative
 - .○ : supplements the narrative

Sprint Planning

Scrum Events

Sprint Planning

Daily Scrum

Sprint Review

Sprint Retrospective

product backlog

↓
sprint backlog

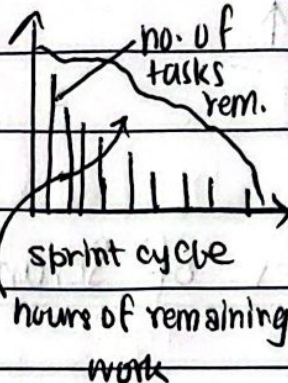
↓
sprint goal

↓
planning poker

Quick Sync,
inspect progress
and adapt plan
for next 24 hrs.

↓
15 mins meeting

↓
impediments
resolved



↓ Show stakeholders
what was "Done".
Get Feedback.

— ~ —
Reflect on how
the Sprint went and
what to
change for next
Sprint

Kanban Board

Backlog

Doing

Review

Done

Item 1

Item 2

Throughput = $\frac{\text{WIPs}}{\text{Cycle Time}}$

avg tasks completed per cycle time

WIP: no. of tasks between
Doing and Done

Cycle Time: ^{avg} time for a
task to be done

Advantages of Scrum:

- speed ↑
- scope-focused & user focused ↑
- teamwork ↑
- flexibility + adaptability
- motivation ↑
- feedback ↑

Disadvantages of Scrum

- documentation can be neglected.
- quality ↓ if pressure leads to shortcuts or 'Done' is rigor.
- requires product owner to manage backlog effectively.
- team challenges → difficult if geographically dispersed
 - requires good communication
 - replacing members mid-stream is disruptive

DevOps

Developers and operations work together from the beginning. So developers must think about deployment, performance, reliability and maintenance. Operations give feedback early.

- decreases deployment time.
- accountability ↑
- small team that can handle the full flow.
- organize work around value streams, sequence off of activities needed to deliver value to the customer.

How DevOps Teams Work

- a) limit multitasking
- b) small batches, small steps.
- c) reduce handoffs.
- d) eliminate waste → work not creating value.
- e) Swarm problems
- f) Quality is built in.

Pros:

- Focus on building resilient systems.
- Integrates security practices early.
- Learns ~~to~~ from failures → debriefing
→ conducts blameless ^{postmortems} ~~not~~ after real incidents.

Comparing SDLC and Agile

Goals	SDLC	Agile
Reduce Interface time & errors	Relies on Standards, common forms, defined procedures, Consistency ↑	Relies on pair programming. Quality through collab.
Reduce Process learning & Dual Processing	manages releases carefully to minimize users learning multi systems at once	rapid development and prototyping. Less emphasis on learning CASE upfront. Faster feedback reduces dual processing needs.
Reduce Task Structuring & Formatting time	Uses CASE Tools, diagrams, templates, code reuses.	Short releases

Reduce Data/ Knowledge Search Time/Cost	Interviews, Observation STROBE, sampling	On-site customer for immediate access to info Risk if customer info is flawed.
reduce comm. & coordination time/cost	breaks project into smaller tasks/teams,	Uses timeboxing to focus communication. Values comm, potentially increasing comm time to increase effectiveness.
Reduce Human Losses from Human Info Overload	May filter info reaching developers. Aims to protect focus	Emphasizes sustainable pace. Avoids rested Assumes rested developers handle info better and produce higher quality work.

Risks Adopting Agile

- Organizational Culture → agile thrives in innovative culture.
- Timing - don't ~~inst~~ introduce major change during instability.
- Cost $\left\{ \begin{array}{l} \text{Training scrum masters, developers, product owners} \\ \text{time spent learning vs doing proj. work.} \end{array} \right.$
- Clients Reaction - will they embrace strong involvement?
- Measuring Impact - Agile's history is shorter than SDLC.
- Individual Rights/Prefs $\left\{ \begin{array}{l} \text{Some devs like working alone} \\ \text{Need to balance methodology with respecting} \\ \text{individual creativity.} \end{array} \right.$